1.Install and run the free version of the OpenVAS or Nessus vulnerability scanner on your system, and perform a basic scan of your own computer or a local virtual machine.

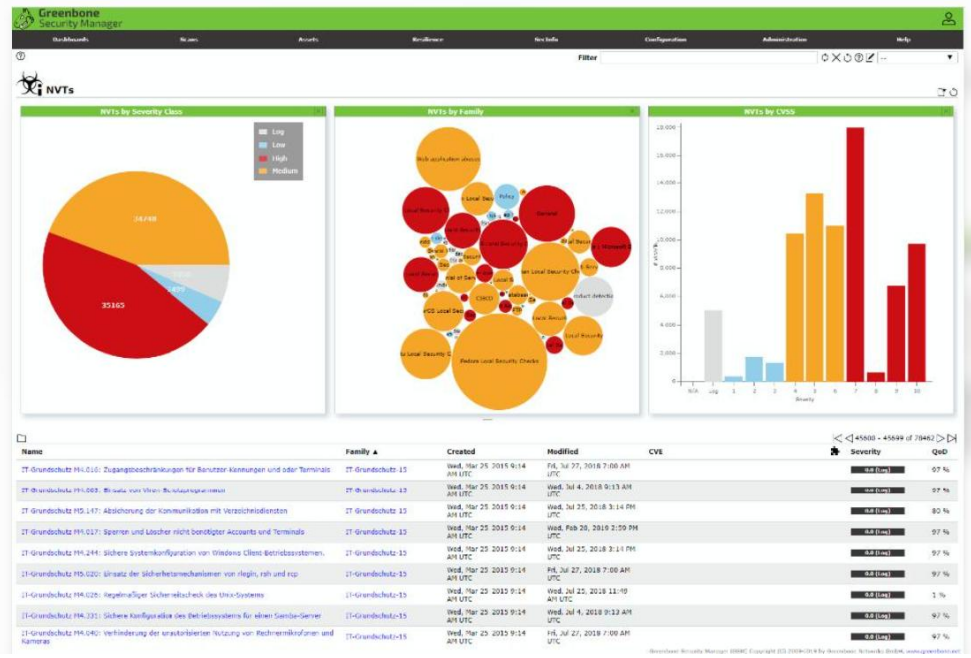
Ans-Step1 :sudo apt update

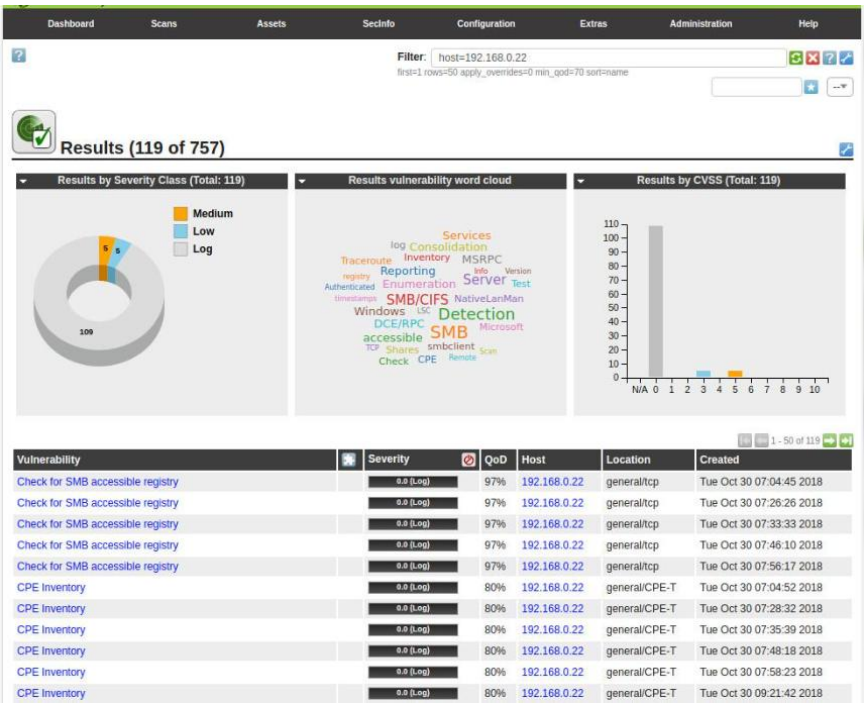
sudo apt upgrade -y

Step 2:sudo apt install gvm -y

Step 3: sudo gvm-setup

Step 4: sudo gvm-start

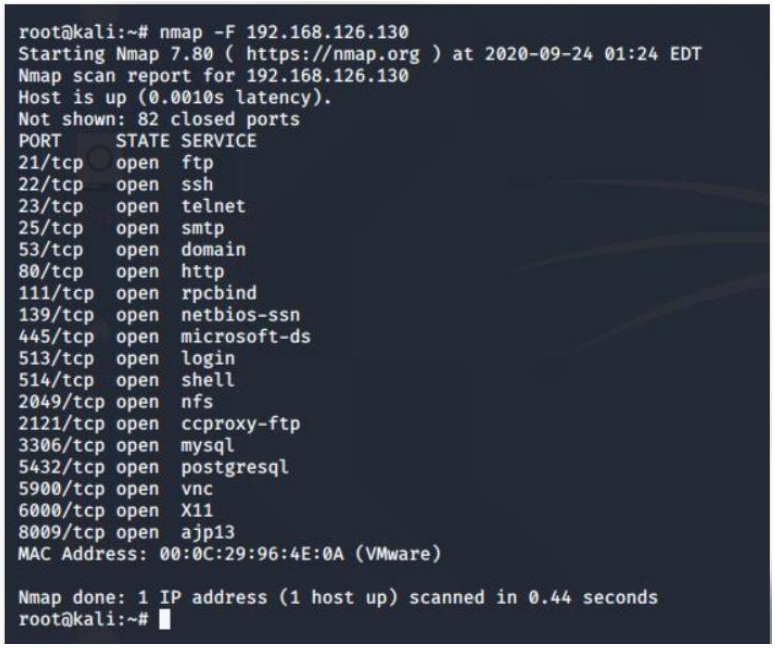




**2.Scan your Wi-Fi router's IP address using Nmap and list all open ports detected.

Hint: Use the command 'nmap -v [router_ip]' in your terminal or command prompt.**

Ans-



3.Analyze the scan report from your vulnerability scanner and identify at least 3 vulnerabilities or warnings it reports; for each, write a one-line description of what the risk is.

Ans-Step 1: Open your Greenbone dashboard

1. Open Scans.
2. Select Reports.
3. Open your completed scan.
4. Check the Results section.
5. Note the severity and vulnerability name.

Step 2: Understand severity levels

Severity	Meaning
Critical	Very serious security risk
High	Major security risk
Medium	Moderate security risk
Low	Minor security risk

Log	Informational finding

Step 3: Write three findings

The following are examples for your assignment. Use the actual findings from your report if they differ.

Vulnerability / Warning	Risk description
Outdated software	Old software may contain known security weaknesses that attackers can exploit.
SSL/TLS configuration weakness	Weak encryption settings may expose communication to interception or downgrade attacks.
Missing security updates	Unpatched systems may be vulnerable to publicly known attacks.

Step 4: Write your practical observation

You can use this format:

Observation: The vulnerability scanner identified security weaknesses and configuration warnings in the scanned system. These findings help administrators understand security risks and take corrective actions**.**

4.Pick any app you use daily (like Instagram, Zomato, or Paytm) and research one real vulnerability that was found in it in the past; write a short summary of what the vulnerability was and how it could have been detected by scanning tools.

Ans-

Objective: Research a real-world vulnerability and explain how security testing could identify similar weaknesses.

Step 1: Research the vulnerability

In July 2019, security researcher Laxman Muthiyah discovered a vulnerability in Instagram's mobile password recovery process.

The vulnerability involved the six-digit password reset verification code.

Instagram had implemented rate limiting to prevent repeated guessing attempts, but the researcher found a way to bypass this protection.

This could potentially allow an attacker to take control of Instagram accounts.

Facebook fixed the vulnerability and awarded the researcher a $30,000 bug bounty.

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Step 2: Understand the risk

Risk: An attacker could potentially bypass password recovery protections and gain unauthorized access to a victim's Instagram account.

Step 3: How could scanning tools detect similar vulnerabilities?

Security testing tools can help identify:

1. Rate-limiting weaknesses: Check whether repeated requests are properly restricted.
2. API security issues: Test whether password reset APIs enforce proper authorization and validation.
3. Authentication weaknesses: Check whether verification codes and password reset processes are securely implemented.

**5.Use ChatGPT or Copilot to generate a basic bash or PowerShell script that runs Nmap on a list of 3 IP addresses and saves the results to separate files; copy and run the script, and share the output files.

Hint: Ask the AI tool for a script that loops through IPs and uses Nmap for each.**

Ans-

